

CSE 3206 – Software Engineering Sessional

Lab 2: Software Process Models, Requirement Analysis & MVP Development

Course Outcome (CO2)

Design and implement software solutions that effectively address user requirements and enhance system reliability and performance.

Program Outcome (PO3)

Design and Development of Software Solutions

Assessment Tool

Project Design Report, GitHub Repository, MVP Demonstration, Individual Viva

Lab Overview

In this lab, each team will work as an independent **software development company**. Your team has been awarded a software development contract from a client. The client has provided a project scenario describing a real-world problem.

Your responsibility is to:

- Analyze the client's requirements.
- Select the most appropriate software development process model.
- Design the proposed software solution.
- Develop a **Minimum Viable Product (MVP)** demonstrating the core functionality.
- Collaborate professionally using Git and GitHub.
- Document your engineering decisions in a Project Design Report.

This is **not a complete software development project**. Instead, it is the first milestone of a semester-long project where each lab builds upon the previous one.

Learning Outcomes

After completing this lab, students will be able to:

- Analyze a real-world software engineering problem.
- Identify stakeholders and user requirements.
- Differentiate among software process models.
- Select and justify an appropriate software development methodology.
- Plan software development collaboratively.
- Develop a small functional prototype (MVP).
- Apply Git and GitHub in collaborative software development.
- Produce professional software engineering documentation.

Project Assignment

- Students will work in **teams of three members**.
- Each team will continue working on the **same project** throughout the semester.
- Teams may use **any programming language, framework, database, IDE, or development platform**.

Laboratory Activities

Phase 1 – Requirement Analysis

Analyze the assigned project and identify the client's needs.

Complete the following:

- Project overview
- Problem statement
- Stakeholder identification
- Project scope
- Functional requirements (minimum 10)
- Non-functional requirements (minimum 8)
- Five user stories or use cases
- Assumptions and constraints

Phase 2 – Software Process Model Selection

Study the following software process models:

- Waterfall
- Incremental
- Prototype
- Spiral
- Rapid Application Development (RAD)
- Agile XP
- Scrum
- Kanban

Based on your project characteristics:

- Select **one** software process model.
- Justify your selection.
- Discuss why the chosen model is suitable.
- Explain why at least **two alternative models** are less suitable.

There is **no single correct answer**. Your reasoning and justification will be evaluated.

Phase 3 – MVP Design and Development

Develop a **Minimum Viable Product (MVP)** that demonstrates the feasibility of your proposed software solution.

The MVP should include only the essential functionalities required to demonstrate the project concept.

Examples include:

- User authentication
- Dashboard
- One or two core modules
- Basic CRUD operations
- Navigation between pages
- Sample data management
- Simple user interface

The MVP should remain **small, functional, and modular**. Teams should focus on applying software engineering principles rather than implementing numerous features.

Phase 4 – GitHub Collaboration

Each team will maintain a shared GitHub repository.

Every team member must contribute individually using Git.

Recommended workflow:

```
main
|
├── member1-feature
├── member2-feature
└── member3-feature
```

Each member must:

- Create a feature branch
- Implement assigned tasks
- Commit changes with meaningful messages
- Push changes to GitHub
- Create Pull Requests
- Review teammates' code
- Resolve conflicts if necessary
- Merge approved changes into the main branch

The repository history should clearly demonstrate each member's contribution.

Development Guidelines

Teams are free to choose:

- Programming language
- Framework
- Database
- IDE
- Architecture
- UI design
- Development tools

Evaluation will emphasize **engineering methodology, teamwork, and design decisions**, rather than the complexity of the implementation.

GitHub Requirements

Each repository should include:

README.md

docs/

 Requirement_Report.pdf

src/

assets/

screenshots/

Minimum GitHub activities:

- Repository creation
- README
- Minimum three feature branches
- Minimum three Pull Requests
- Branch merging
- Meaningful commit history

Project Design Report Structure

1. Cover Page
2. Team Information
3. Project Title
4. Problem Statement
5. Project Objectives
6. Stakeholder Analysis
7. Functional Requirements
8. Non-functional Requirements
9. User Stories / Use Cases
10. Selected Software Process Model
11. Justification of Process Model
12. Comparison with Alternative Models
13. MVP Design Overview
14. GitHub Collaboration Evidence
15. Challenges Encountered
16. Conclusion

Expected Deliverables

Each team will submit:

1. GitHub Repository

Including:

- Source code
- Branches
- Pull Requests
- Commit history
- README

2. Project Design Report (PDF)

Prepared according to the specified format.

3. MVP Demonstration

A working prototype demonstrating the core functionalities.

Important Guidelines

- This is the **first milestone** of a semester-long software development project.
- Develop only a **Minimum Viable Product (MVP)**; a complete software system is **not** expected.
- Organize the project into clear, reusable modules to support future labs on design patterns and testing.
- Every team member must contribute through GitHub using feature branches and pull requests.
- Creativity, sound engineering decisions, collaboration, and documentation will be valued more than the number of implemented features.
- Teams are encouraged to follow professional software engineering practices while maintaining a manageable project scope.

60 Different Project Scenarios:

Group	Project	Suggested Best Process Model	Team name
1	University Course Registration System	Agile Scrum	Group#01 from Section A (1 st 30)
2	Online Book Store	Incremental	Group#02 from Section A (1 st 30)
3	Hospital Appointment System	Spiral	Group#03 from Section A (1 st 30)
4	Restaurant Ordering System	Prototype	Group#04 from Section A (1 st 30)
5	Smart Parking System	Agile XP	Group#05 from Section A (1 st 30)
6	Pharmacy Management System	Waterfall	Group#06 from Section A (1 st 30)
7	Blood Donation Management System	Agile Scrum	Group#07 from Section A (1 st 30)
8	Railway Ticket Booking System	Incremental	Group#08 from Section A (1 st 30)
9	Hotel Reservation System	RAD	Group#09 from Section A (1 st 30)
10	School Management System	Waterfall	Group#10 from Section A (1 st 30)
11	Food Delivery Application	Kanban	Group#01 from Section A (2 nd 30)

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

#Lab 2

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Marks: 10

12	E-Learning Platform	Agile Scrum	Group#02 from Section A (2 nd 30)
13	Library Management System	Waterfall	Group#03 from Section A (2 nd 30)
14	Vehicle Rental System	Prototype	Group#04 from Section A (2 nd 30)
15	Online Examination System	Spiral	Group#05 from Section A (2 nd 30)
16	Digital Wallet System	Agile XP	Group#06 from Section A (2 nd 30)
17	Gym Membership Management	RAD	Group#07 from Section A (2 nd 30)
18	Super Shop POS System	Incremental	Group#08 from Section A (2 nd 30)
19	Smart Waste Collection System	Agile Scrum	Group#09 from Section A (2 nd 30)
20	Online Doctor Consultation	Spiral	Group#10 from Section A (2 nd 30)
21	Insurance Claim System	Waterfall	Group#01 from Section B (1 st 30)
22	Courier Management System	Kanban	Group#02 from Section B (1 st 30)
23	Smart Irrigation System	Prototype	Group#03 from Section B (1 st 30)
24	Bus Ticket Reservation	Incremental	Group#04 from Section B (1 st 30)
25	University Hostel Management	Waterfall	Group#05 from Section B (1 st 30)
26	Disaster Relief Coordination System	Agile Scrum	Group#06 from Section B (1 st 30)
27	Airport Baggage Tracking System	Spiral	Group#07 from Section B (1 st 30)
28	Digital Banking System	Waterfall	Group#08 from Section B (1 st 30)
29	Online Freelancing Marketplace	Agile XP	Group#09 from Section B (1 st 30)
30	Inventory Management System	Incremental	Group#10 from Section B (1 st 30)
31	Event Management System	RAD	Group#01 from Section B (2 nd 30)
32	Smart Home Automation	Prototype	Group#02 from Section B (2 nd 30)
33	Online Voting System	Spiral	Group#03 from Section B (2 nd 30)
34	Clinic Management System	Agile Scrum	Group#04 from Section B (2 nd 30)
35	Agricultural Product Marketplace	Kanban	Group#05 from Section B (2 nd 30)
36	Tourism Management System	Incremental	Group#06 from Section B (2 nd 30)
37	Pet Care Management System	Prototype	Group#07 from Section B (2 nd 30)
38	Ride Sharing System	Agile Scrum	Group#08 from Section B (2 nd 30)
39	Employee Attendance System	Waterfall	Group#09 from Section B (2 nd 30)
40	NGO Donation Management System	RAD	Group#10 from Section B (2 nd 30)

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41	Online Grocery Delivery System	Agile Scrum	Group#01 from Section C (1 st 30)
42	Car Workshop Management System	Incremental	Group#02 from Section C (1 st 30)
43	Online Job Portal	Spiral	Group#03 from Section C (1 st 30)
44	Real Estate Listing Platform	Prototype	Group#04 from Section C (1 st 30)
45	Fitness Tracker Application	Agile XP	Group#05 from Section C (1 st 30)
46	Municipal Waste Billing System	Waterfall	Group#06 from Section C (1 st 30)
47	Online Tutoring Platform	Agile Scrum	Group#07 from Section C (1 st 30)
48	Cargo Tracking System	Incremental	Group#08 from Section C (1 st 30)
49	Wedding Planning Management System	Kanban	Group#09 from Section C (1 st 30)
50	Public Library Digital Catalog System	Waterfall	Group#10 from Section C (1 st 30)
51	Online Pharmacy Delivery System	Kanban	Group#01 from Section C (2 nd 30)
52	Virtual Classroom Management System	Agile Scrum	Group#02 from Section C (2 nd 30)
53	Vehicle Maintenance Tracker	Waterfall	Group#03 from Section C (2 nd 30)
54	Car Pooling System	Prototype	Group#04 from Section C (2 nd 30)
55	Online Auction System	Spiral	Group#05 from Section C (2 nd 30)
56	Smart Traffic Management System	Agile XP	Group#06 from Section C (2 nd 30)
57	Co-working Space Booking System	Prototype	Group#07 from Section C (2 nd 30)
58	Restaurant Inventory & Supply Management	Incremental	Group#08 from Section C (2 nd 30)
59	Online Charity Crowdfunding Platform	Agile Scrum	Group#09 from Section C (2 nd 30)
60	Municipal Complaint Management System	Spiral	Group#10 from Section C (2 nd 30)

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Rubric for Lab 2: Exploring Software Process Models & Requirement Analysis

Course Outcome (CO2) – Program Outcome (PO3)

Criteria	Excellent (100%)	Good (80%)	Satisfactory (60%)	Needs Improvement (40%)	Marks
Requirement Analysis	Clearly identifies stakeholders, project scope, and comprehensive functional & non-functional requirements	Minor omissions in requirements or stakeholder analysis	Basic requirements identified with limited analysis	Requirements incomplete or unclear	3
Software Process Model Selection & Justification	Appropriate process model selected with strong justification and comparison with alternatives	Appropriate model selected with adequate justification	Model selected with limited justification	Model selection inappropriate or lacks justification	2
Mini Project (MVP) Design & Implementation	Functional MVP demonstrating core requirements with good design and organization	MVP implemented with minor missing features	Basic prototype developed with limited functionality	Prototype incomplete or non-functional	2
GitHub Collaboration	Proper repository, individual branches, meaningful commits, pull requests, and successful merge by all members	Repository and branches maintained with minor issues	Limited GitHub collaboration; few commits or branches	Little or no evidence of collaborative GitHub workflow	2
Project Design Report & Presentation	Well-organized report with complete documentation and professional presentation	Report mostly complete with minor deficiencies	Basic report with limited explanation	Poorly organized or incomplete report	1
Total					10

Performance Level Guide

- **Excellent:** 90–100% of criterion achieved
- **Good:** 75–89% of criterion achieved
- **Satisfactory:** 60–74% of criterion achieved
- **Needs Improvement:** Below 60% of criterion achieved